Programming Fundamentals Lab 11

Lab Tasks 3, 5-10

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Course: Programming Fundamentals Lab
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Note: All of the Tasks are available on my Github.

Task 3 (from previous 4 tasks):

```
#include <stdio.h>
#include <string.h>
struct Flight
{
	int flight_number;
	char departure_city[30];
	char destination_city[30];
	char date[12];
	int avaliable_seats;
};
void bookseat(struct Flight tickets[5], int *current);
void addflight(struct Flight tickets[5], int *current);
void allflights(struct Flight tickets[5], int *current);
int main()
{
	struct Flight tickets[5];
	int choice, current=0;
	do
	{
		printf("\n\nWelcome to Fight System.");
		printf("\n 1. Add a flight.");
		printf("\n 2. Book a seat.");
		printf("\n 3. Show all flights.");
		printf("\n 4. Exit the system.");
		printf("\nEnter your choice.\n");
```

```
		scanf("%d", &choice);
		switch (choice)
		{
		case 1:
			addflight(tickets, &current);
			break;
		case 2:
			bookseat(tickets, &current);
			break;
		case 3:
			allflights(tickets, &current);
			break;
		case 4:
			break;
		default:
			printf("\nInvalid Choice.\n");
			break;
		}
	} while (choice!=4);
	printf("\n");
return 0;
}

void addflight(struct Flight tickets[5], int *current)
{
	printf("\nEnter your flight number: ");
```

```
	scanf("%d", &tickets[*current].flight_number);

	printf("\nEnter your departure city: ");
	getchar();
	fgets(tickets[*current].departure_city, sizeof(tickets[*current].departure_city), stdin);
	tickets[*current].departure_city[strcspn(tickets[*current].departure_city, "\n")]='\0';

	printf("\nEnter your destination city: ");
	fgets(tickets[*current].destination_city, sizeof(tickets[*current].destination_city), stdin);
	tickets[*current].destination_city[strcspn(tickets[*current].destination_city, "\n")]='\0';

	printf("\nEnter your flight date: ");
	fgets(tickets[*current].date, sizeof(tickets[*current].date), stdin);
	tickets[*current].date[strcspn(tickets[*current].date, "\n")]='\0';

	printf("\nEnter the number of avaliable seats: ");
	scanf("%d", &tickets[*current].avaliable_seats);

	(*current)++;
}

void bookseat(struct Flight tickets[5], int *current)
{
	int tempflight, seats;
	printf("\nEnter the flight number of the flight: ");
	scanf("%d", &tempflight);
```

```
	for (int i = 0; i < (*current); i++)
	{
		if (tickets[i].flight_number == tempflight)
		{
			printf("\nThere are %d seats avaliable, Enter the number of seats to book: ", tickets[i].avaliable_seats);
			scanf("%d", &seats);
			if (seats <= tickets[i].avaliable_seats)
			{
				tickets[i].avaliable_seats-=seats;
				printf("\nSeats have been booked.\n");
				return;
			}
			else
			{
				printf("\nNot enough seats avaliable.\n");
				return;
			}
		}
	}
	printf("\nInvalid flight number.\n");
	return;
}

void allflights(struct Flight tickets[5], int *current)
{
```

```
	printf("\nAll Current Flights\n");
	for (int i = 0; i < (*current); ++i)
	{
		printf("\nFlight Number: %d", tickets[i].flight_number);
		printf("\nDeparture City: %s", tickets[i].departure_city);
		printf("\nDestination City: %s", tickets[i].destination_city);
		printf("\nDate of Flight: %s", tickets[i].date);
		printf("\nAvaliable Seats: %d", tickets[i].avaliable_seats);
		printf("\n");
	}
}
```

```
Welcome to Fight System.
 1. Add a flight.
 2. Book a seat.
 3. Show all flights.
 4. Exit the system.
Enter your choice.
1

Enter your flight number: 123

Enter your departure city: New York

Enter your destination city: Karachi

Enter your flight date: 2-11-2025

Enter the number of avaliable seats: 10


Welcome to Fight System.
 1. Add a flight.
 2. Book a seat.
 3. Show all flights.
 4. Exit the system.
Enter your choice.
2

Enter the flight number of the flight: 123

There are 10 seats avaliable, Enter the number of seats to book: 5

Seats have been booked.


Welcome to Fight System.
 1. Add a flight.
 2. Book a seat.
 3. Show all flights.
 4. Exit the system.
Enter your choice.
3

All Current Flights

Flight Number: 123
Departure City: New York
Destination City: Karachi
Date of Flight: 2-11-2025
Avaliable Seats: 5
```

```
All Current Flights

Flight Number: 123
Departure City: New York
Destination City: Karachi
Date of Flight: 2-11-2025
Avaliable Seats: 5


Welcome to Fight System.
 1. Add a flight.
 2. Book a seat.
 3. Show all flights.
 4. Exit the system.
Enter your choice.
4

Press any key to continue . . .
```

Task 5:

```
#include <stdio.h>

#include <string.h>

void reverse(char message[100], int len);

int main()

{

	char message[100];

	int len;

	printf("Enter your secret message:\n");

	fgets(message, sizeof(message), stdin);

	len = strlen(message);

	reverse(message, len);

	printf("\n");

	return 0;

}


void reverse(char message[100], int len)

{

	if (len<0)

	{

		return;

	}

	printf("%c", message[len]);

	reverse(message, len-1);

}
```

```
Enter your secret message:
Hello Sir Sheeraz

zareehS riS olleH
Press any key to continue . . .
```

Task 6:

```
#include <stdio.h>

int expo(int num, int power);

int main()

{

	int num, power, final;

	printf("Enter the Number: ");

	scanf("%d", &num);

	printf("\nEnter the exponent: ");

	scanf("%d", &power);

	final = expo(num, power);

	printf("\nThe number is %d", final);

	printf("\n");

	return 0;

}


int expo(int num, int power)

{

	if (power==0)~~

	{

		return 1;

	}

	power--;

	return num * expo(num,power);

}
```

```
Enter the Number: 4

Enter the exponent: 4

The number is 256
Press any key to continue . . .
```

Task 7:

```
#include <stdio.h>
#include <string.h>
struct department
{
	char name[50];
	int blocknumber;
};
struct student
{
	int id;
	char name[50];
	float cgpa;
	struct department dept;
};

int main()
{
	struct student stud;
	printf("Enter Student ID: ");
	scanf("%d", &stud.id);
	printf("\nEnter Student Name: ");
	getchar();
	fgets(stud.name, sizeof(stud.name), stdin);
	printf("\nEnter Student CGPA: ");
	scanf("%f", &stud.cgpa);
```

```
    printf("\nEnter Student department: ");
    getchar();
    fgets(stud.dept.name, sizeof(stud.dept.name), stdin);
    printf("\nEnter Department Block Number: ");
    scanf("%d", &stud.dept.blocknumber);

    printf("\n\nStudent Data:");
    printf("\nStudent ID:\t %d", stud.id);
    printf("\nStudent Name:\t %s", stud.name);
    printf("\nStudent CGPA:\t %.3f", stud.cgpa);
    printf("\nDepartment:\t %s", stud.dept.name);
    printf("\nDepartment Block:   %d\n", stud.dept.blocknumber);
    return 0;
}
```

```
Enter Student ID: 0538

Enter Student Name: Farhan

Enter Student CGPA: 3.4

Enter Student department: Computer Science

Enter Department Block Number: 2


Student Data:
Student ID:      538
Student Name:    Farhan

Student CGPA:    3.400
Department:      Computer Science

Department Block:   2
Press any key to continue . . .
```

Task 8:

```
#include <stdio.h>

#include <string.h>


struct employee

{

	int id;

	char name[50];

	int salary;

	char position[50];

};


int main()

{

	struct employee emps[5];

	int i, highest=0, num=0;

	for (i = 0; i < 5; ++i)

	{

	printf("Enter employee#%d ID: ", i+1);

	scanf("%d", &emps[i].id);

	printf("\nEnter employee#%d name: ", i+1);

	getchar();

	fgets(emps[i].name, sizeof(emps[i].name), stdin);

	printf("\nEnter employee#%d salary: ", i+1);

	scanf("%d", &emps[i].salary);

	if (emps[i].salary > highest)
```

```
	{
		highest=emps[i].salary;
		num=i;
	}
	printf("\nEnter employee#%d position: ", i+1);
	getchar();
	fgets(emps[i].position, sizeof(emps[i].position), stdin);
	}
	printf("\n\nThe highest salary belongs to:");
	printf("employee ID:\t%d\nEmployee Name:\t%sSalary:\t%d\nPosition:\t%s\n",
emps[num].id,emps[num].name,emps[num].salary,emps[num].position);
	return 0;
}
```

```
Enter employee#1 ID: 1

Enter employee#1 name: Farhan

Enter employee#1 salary: 100000

Enter employee#1 position: Cyber Security Expert
Enter employee#2 ID: 2

Enter employee#2 name: Jawad

Enter employee#2 salary: 80000

Enter employee#2 position: Data Analyist
Enter employee#3 ID: 3

Enter employee#3 name: Zohaib

Enter employee#3 salary: 150000

Enter employee#3 position: Facility Doctor
Enter employee#4 ID: 4

Enter employee#4 name: Usman

Enter employee#4 salary: 90000

Enter employee#4 position: Electrical Engineer
Enter employee#5 ID: 5

Enter employee#5 name: Saqlain

Enter employee#5 salary: 300000

Enter employee#5 position: Project Manager


The highest salary belongs to:employee ID:      5
Employee Name:  Saqlain
Salary: 300000
Position:       Project Manager

Press any key to continue . . .
```

Task 9:

```
#include <stdio.h>
void check();
int main()
{
	int choice;
	do
	{
		printf("\nPress 1 to call function, any other key to exit.\n");
		scanf("%d", &choice);
		if (choice==1)
		{
			check();
		}
	} while (choice == 1);

	return 0;
}

void check()
{
	static int a=1;
	printf("\nThis Function has been called %d times.\n", a);
	a++;
}
```

```
Press 1 to call function, any other key to exit.
1

This Function has been called 1 times.

Press 1 to call function, any other key to exit.
1

This Function has been called 2 times.

Press 1 to call function, any other key to exit.
1

This Function has been called 3 times.

Press 1 to call function, any other key to exit.
1

This Function has been called 4 times.

Press 1 to call function, any other key to exit.
1

This Function has been called 5 times.

Press 1 to call function, any other key to exit.
2
Press any key to continue . . .
```

Task 10:

```
#include <stdio.h>
#include <string.h>
struct director
{
	char name[50];
	char nationality[50];
};
struct movies
{
	char title[50];
	char genre[50];
	struct director direc;
	int release_year;
	float rating;
};
void addmovie(struct movies moveee[10], int *currentmoveee);
void searchgenre(struct movies moveee[10], int currentmoveee);
void display(struct movies moveee[10], int currentmoveee);
int main()
{
	struct movies moveee[10];
	int currentmoveee=0;
	int choice;
	do
```

```
	{
		printf("\n\n1. Add a new movie.\n2. Search a movie by genre.\n3. Display all movies.\n4. Exit program.\nEnter your choice.\n");
		scanf("%d", &choice);
		switch (choice)
		{
		case 1:
			addmovie(moveee, &currentmoveee);
			printf("\nMovie successfully added.\n");
			break;
		case 2:
			searchgenre(moveee, currentmoveee);
			break;
		case 3:
			display(moveee, currentmoveee);
			break;
		case 4:
			break;
		default:
			printf("\nInvalid Choice.\n");
		}
	} while(choice!=4);
	printf("\n");
return 0;
}
```

```
void addmovie(struct movies moveee[10], int *currentmoveee)
{
	printf("\nEnter movie name: ");
	getchar();
	fgets(moveee[*currentmoveee].title, sizeof(moveee[*currentmoveee].title), stdin);
	printf("\nEnter the movie genre: ");
	fgets(moveee[*currentmoveee].genre, sizeof(moveee[*currentmoveee].genre), stdin);
	printf("\nEnter the Director's Name: ");
	fgets(moveee[*currentmoveee].direc.name,
sizeof(moveee[*currentmoveee].direc.name), stdin);
	printf("\nEnter the Director's nationality: ");
	fgets(moveee[*currentmoveee].direc.nationality,
sizeof(moveee[*currentmoveee].direc.nationality), stdin);
	printf("\nEnter release year: ");
	scanf("%d", &moveee[*currentmoveee].release_year);
	printf("\nEnter Movie's rating: ");
	scanf("%f", &moveee[*currentmoveee].rating);
	(*currentmoveee)++;
}

void searchgenre(struct movies moveee[10], int currentmoveee)
{
	char searching[50];
	int flag=0;
	printf("\nEnter the exact genre to search: ");
	getchar();
	fgets(searching, 50, stdin);
```

```
	for (int i = 0; i < (currentmoveee); ++i)
	{
		if (strcmp(moveee[i].genre, searching)==0)
		{
			printf("\nMovie Title:\t%s", moveee[i].title);
			printf("Movie Genre:\t%s", moveee[i].genre);
			printf("Movie Director:\t%s", moveee[i].direc.name);
			printf("Movie Director nationality:\t%s", moveee[i].direc.nationality);
			printf("Release Year:\t%d", moveee[i].release_year);
			printf("\nMovie Rating:\t%.1f", moveee[i].rating);
			flag=1;
		}
	}
	if (flag==0)
	{
		printf("\nInvalid Genre or Movie not found.\n");
	}
}

void display(struct movies moveee[10], int currentmoveee)
{
	for (int i = 0; i < currentmoveee; ++i)
	{
		printf("\n\nMovie#%d:", i+1);
		printf("\nMovie Title:\t%s", moveee[i].title);
		printf("Movie Genre:\t%s", moveee[i].genre);
```

```
            printf("Movie Director:\t%s", moveee[i].direc.name);
            printf("Movie Director nationality:\t%s", moveee[i].direc.nationality);
            printf("Release Year:\t%d", moveee[i].release_year);
            printf("\nMovie Rating:\t%.1f", moveee[i].rating);
        }
}
```

```
1. Add a new movie.
2. Search a movie by genre.
3. Display all movies.
4. Exit program.
Enter your choice.
1

Enter movie name: Moonfall

Enter the movie genre: Sci-Fi

Enter the Director's Name: Roland Emmerach

Enter the Director's nationality: German-American

Enter release year: 2022

Enter Movie's rating: 5.2

Movie successfully added.


1. Add a new movie.
2. Search a movie by genre.
3. Display all movies.
4. Exit program.
Enter your choice.
1

Enter movie name: Tarot

Enter the movie genre: Horror

Enter the Director's Name: Spenser Cohen

Enter the Director's nationality: American

Enter release year: 2024

Enter Movie's rating: 4.8

Movie successfully added.
```

```
Movie successfully added.


1. Add a new movie.
2. Search a movie by genre.
3. Display all movies.
4. Exit program.
Enter your choice.
2

Enter the exact genre to search: Horror

Movie Title:    Tarot
Movie Genre:    Horror
Movie Director: Spenser Cohen
Movie Director nationality:     American
Release Year:   2024
Movie Rating:   4.8

1. Add a new movie.
2. Search a movie by genre.
3. Display all movies.
4. Exit program.
Enter your choice.
3


Movie#1:
Movie Title:    Moonfall
Movie Genre:    Sci-Fi
Movie Director: Roland Emmerach
Movie Director nationality:     German-American
Release Year:   2022
Movie Rating:   5.2

Movie#2:
Movie Title:    Tarot
Movie Genre:    Horror
Movie Director: Spenser Cohen
Movie Director nationality:     American
Release Year:   2024
Movie Rating:   4.8

1. Add a new movie.
2. Search a movie by genre.
3. Display all movies.
4. Exit program.
Enter your choice.
4

Press any key to continue . . .
```